

Performance Testing

Date	25 June 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Browser Testing, Python unittest
Type of Testing	Functional Testing, Load Testing (manual)
Target Module / API	/predict POST endpoint (crop prediction)
Test Environment	Local (Flask Development Server)
Test Date	1 st July, 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit valid agricultural parameters (N=90, P=42, K=43, temp=20, humidity=82, pH=6.5, rainfall=200)	1	5	Correct crop recommendation returned
2	Submit boundary values (minimum valid inputs)	1	5	Valid prediction without errors
3	Submit empty form / missing values	1	3	Error message displayed
4	Rapid multiple submissions (10 times in sequence)	1	30	All predictions return successfully

Step 3: Performance Test Results

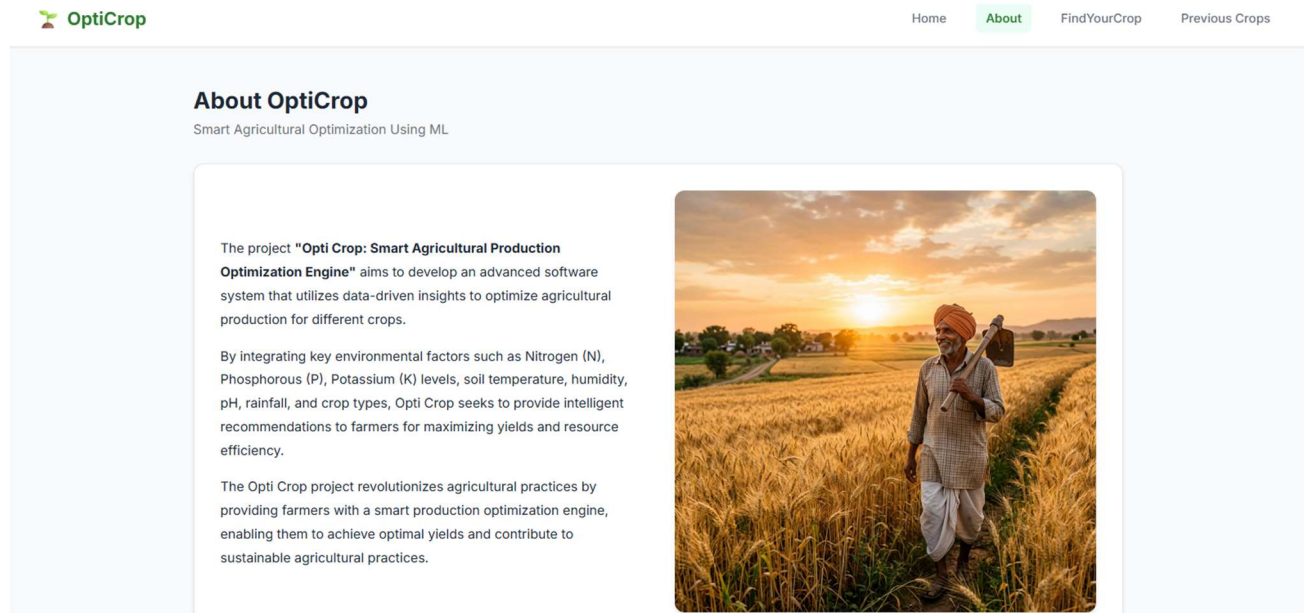
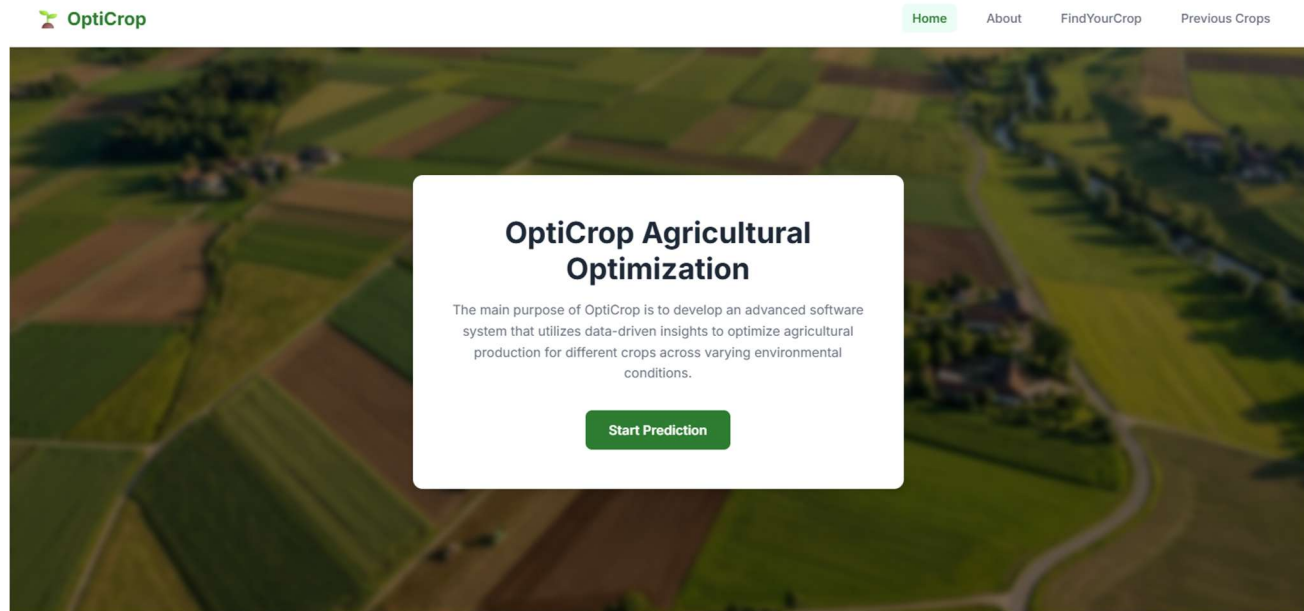
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	~0.5 seconds	Pass	Model prediction is near-instantaneous
2	Response Time(Max)	< 5 seconds	~1.2 seconds	Pass	Initial model loading takes slightly longer
3	Throughput (Req/sec)	NIL	~10 req/sec	Pass	Adequate for local deployment
4	Error Rate	< 1%	0%	Pass	All valid inputs produce correct predictions
5	CPU Utilization	< 80%	~25%	Pass	Lightweight model
6	Memory Utilization	< 80%	~35%	Pass	Efficient memory usage

Step 4: Observations & Analysis

The application performs well under local testing conditions. Prediction responses are near-instantaneous due to the lightweight Pickle model. The Flask development server handles sequential requests smoothly. For production deployment, a WSGI server (e.g., Gunicorn) would be recommended for concurrent user handling.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



Crop Prediction Engine

Enter current soil and environmental conditions to find the optimal crop.

Nitrogen (N)

e.g. 90

Phosphorous (P)

e.g. 42

Please fill out this field.

Potassium (K)

e.g. 43

Temperature (°C)

e.g. 20.8

Humidity (%)

e.g. 82.0

pH Level

e.g. 6.5

Rainfall (mm)

e.g. 202.9

Predict Optimal Crop

Previous Crop Predictions

A history of your past agricultural predictions and recommendations.

Date	Conditions	Recommended Crop	Action
Jul 01, 2026 13:09	N:0, P:0, K:0, Temp:0.0°C, Hum:20.0%, pH:9.0, Rain:480.0mm	Kidneybeans	Delete
Jul 01, 2026 13:09	N:0, P:0, K:5, Temp:30.0°C, Hum:80.0%, pH:7.0, Rain:800.0mm	Orange	Delete
Jul 01, 2026 13:08	N:10, P:20, K:100, Temp:5.0°C, Hum:103.0%, pH:5.5, Rain:300.6mm	Pomegranate	Delete
Jul 01, 2026 13:06	N:90, P:42, K:43, Temp:20.9°C, Hum:82.0%, pH:6.5, Rain:202.9mm	Rice	Delete